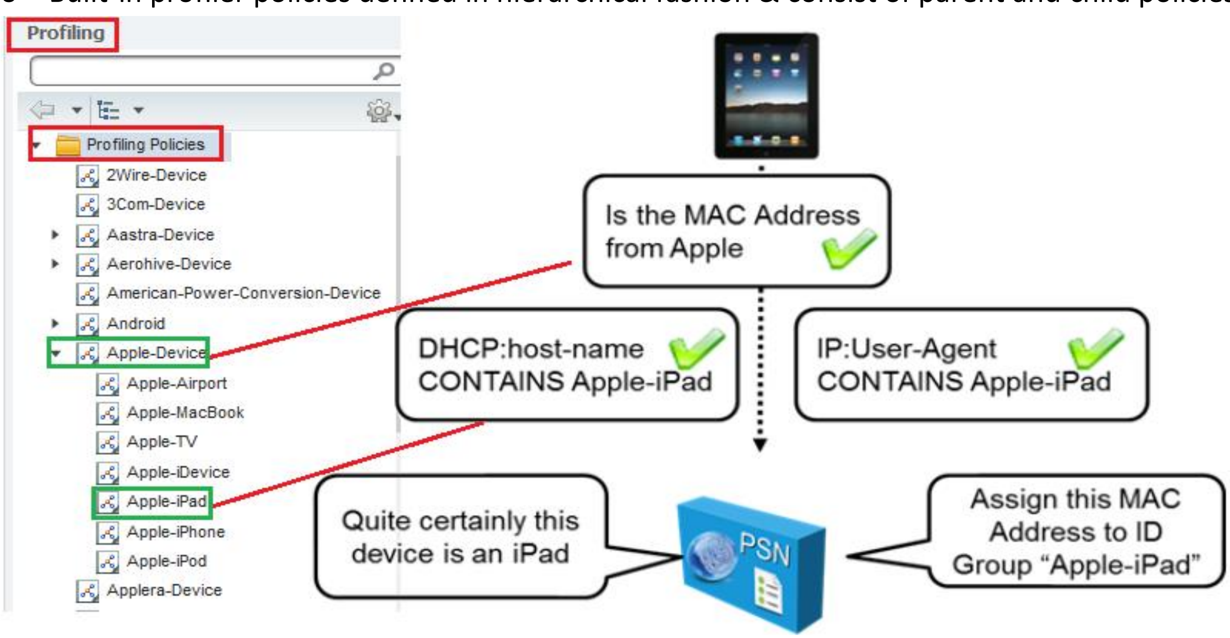
Cisco ISE Profiler:

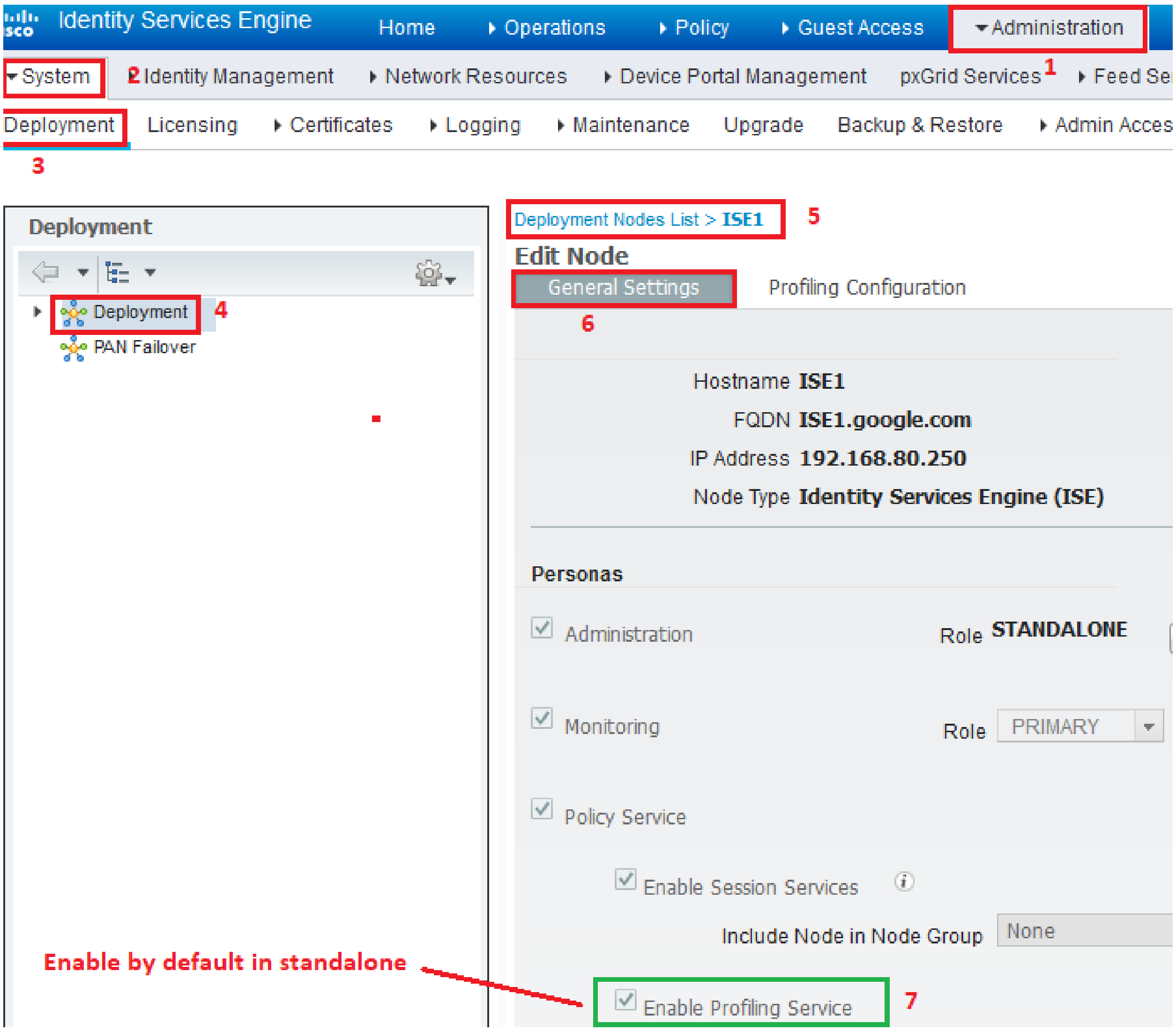
- Profiling Services provides dynamic detection & classification of endpoints.
- Profiler is responsible for endpoint detection & classification connected to network.
- Profiling service identifies devices that connect to network and their location.
- Profiler Identifies locates & determines capabilities of attached network endpoints.
- Profiler using a probe or probes that collect attributes about an endpoint.
- Profiler compares collected attributes to predefined device signatures to locate a match.
- These predefined devices signatures are commonly, referred to as profiles.
- Profiler facilitates effective deployment & ongoing management of authentication.
- Endpoints are profiled, based on the endpoint profiling policies configured in Cisco ISE.
- Profiling feature is very important in authentication & authorization of network access.
- Profiling collects data of Endpoint devices accessing network & identifies its location.
- Depend upon Endpoint profile; authentication server permits access of the resources.
- Profiling service can identify, locate, and ensure the access of all endpoints connected.
- Profiling collects attribute & classifies these endpoints into groups & stores in database.
- Two main functions of profiler service is create and maintain learned inventory.
- Profiler service assign endpoint identity group used in authorization policy.
- Profiler service classify endpoints using variable sets of attributes like MAC address.
- Profiler service using many other attributes to identify and classification endpoints.
- Profiler service discovers & determines capabilities of endpoints attached to network.
- Profiling requires Change of Authority to change the authorization state of an endpoint
- Profiling policies allow categorizing discovered endpoints on network assign to groups.
- Cisco Identify Service Engine comes with numerous built-in profiler conditions and policies.
- Built-in profiler policies defined in hierarchical fashion & consist of parent and child policies.



Implement Profiler Services:

Enable the profiler service on a node. Navigate to **Administration > System > Deployment > General Settings** and check **Enable Profiling Services**

This service is enabled by default and is not configurable when in standalone mode.

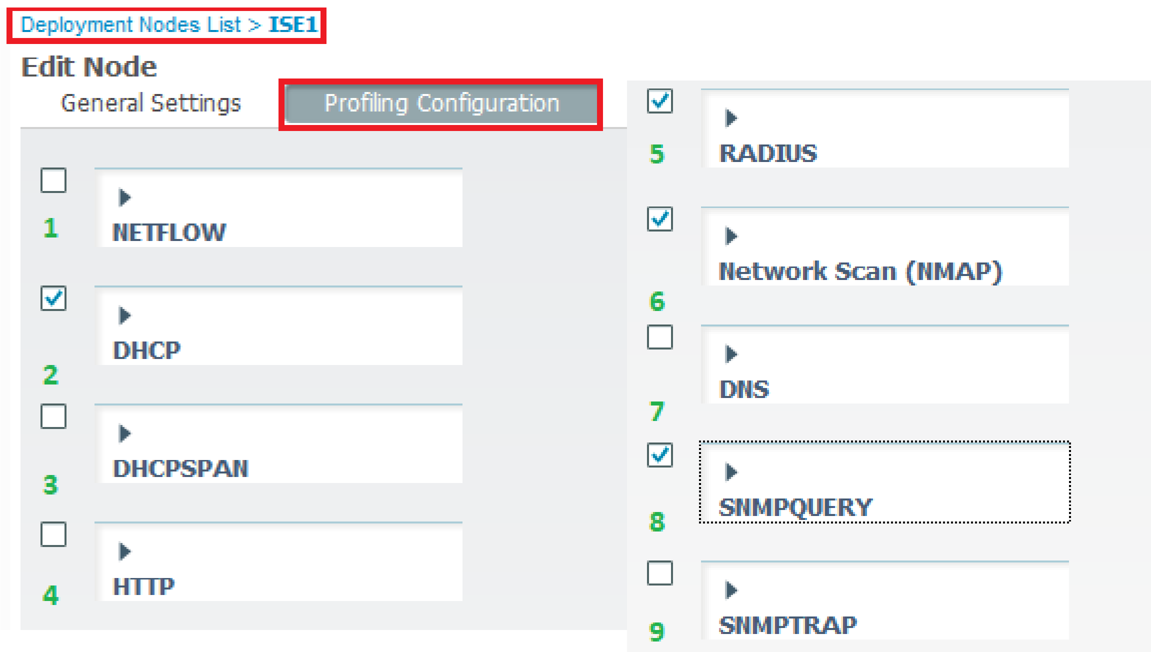


Implement Probes:

- o Probe is a method used to collect attribute or a set of attributes from endpoint on network.
- o By the help of Probe, Profile service collects an attribute or attributes of any endpoint.
- o Probe is software designed to collect data to be used in a profiling decision.
- o By the help of Probe, Profile service create update or modify profile in database.
- o Different Probes are responsible for collection of different type of Endpoint attributes.
- o There are nine probes on each Policy Service Node NETFLOW, DHCP, DHCPSPAN, HTTP,
- o RADIUS, NETWORK SCAN (NMAP), DNS, SNMPQUERY and SNMPTRAP.

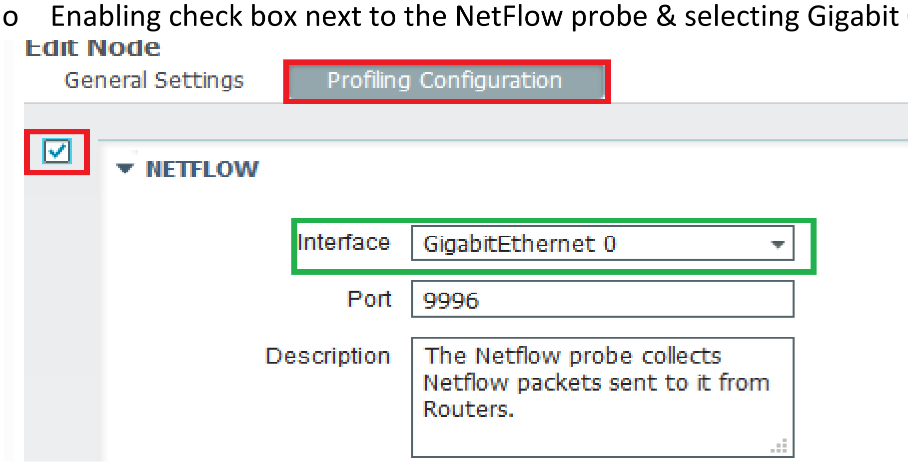
Navigate to **Administration > System > Deployment >** Select the **Profiling Configuration** tab

Click the check box next to the probes want to enable.



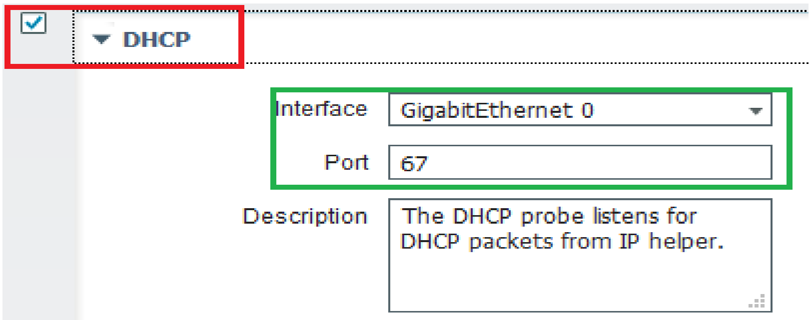
NetFlow Probe:

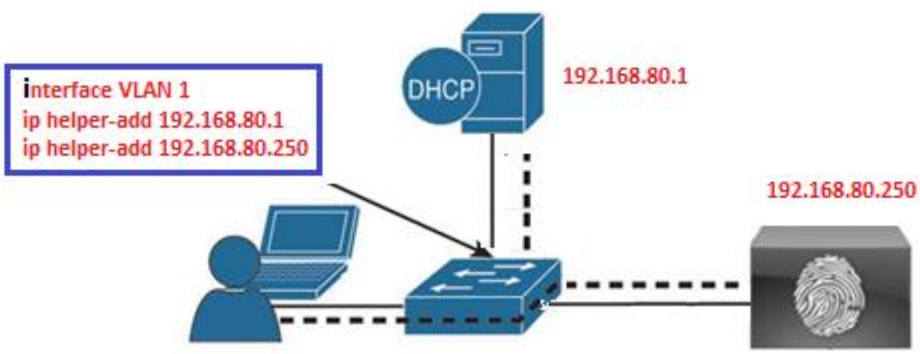
- o Just enabling NetFlow in infrastructure and forwarding it all to the ISE.
- o It is recommended to perform extensive planning prior to use NetFlow probe.
- o Enabling check box next to the NetFlow probe & selecting Gigabit 0 interface.



DHCP Probe:

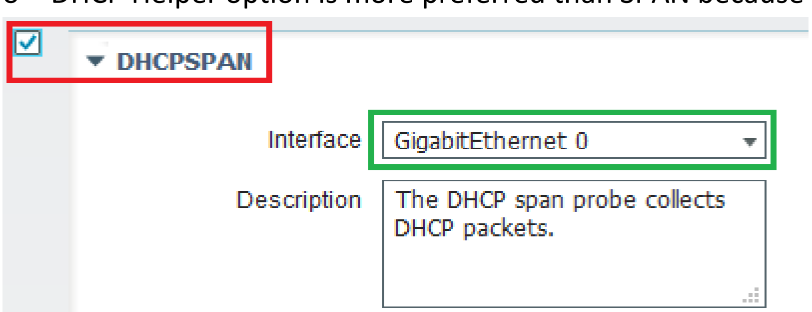
- o Configuring DHCP Packets forwarding directly to the Cisco ISE.
- o Primary use of DHCP in profiling is to capture the device MAC address.
- o DHCP requests also carry User-Agent field that helps to identify OS of device.
- o The DHCP probe requires the DHCP requests to be sent directly to the ISE.
- o Using ip helper-address interface configuration command to send request to ISE.

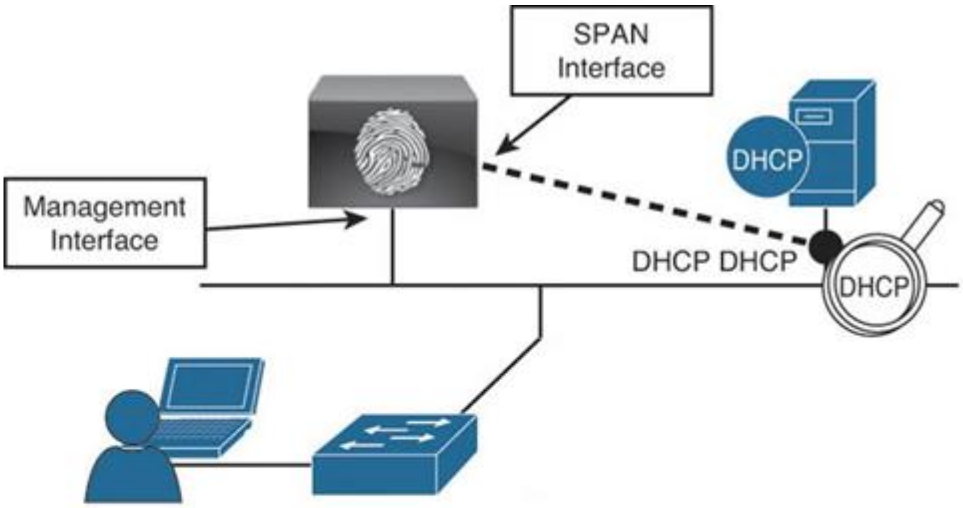




DHCP SPAN Probe:

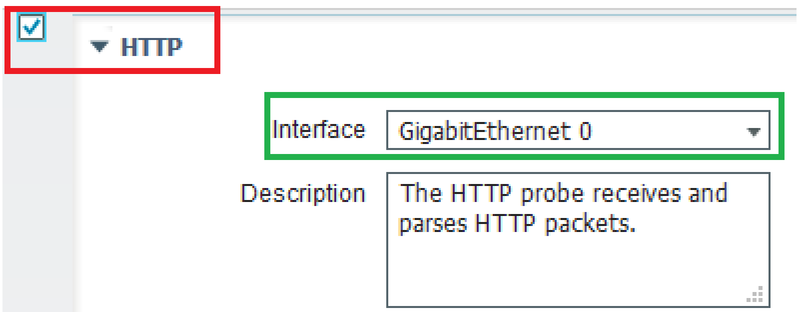
- o DHCP SPAN Probe is not possible or required DHCP Relay agent to configure.
- o SPAN session copies all traffic to/from source interface on a switch to a destination.
- o DHCP Helper option is more preferred than SPAN because it has less traffic overhead.

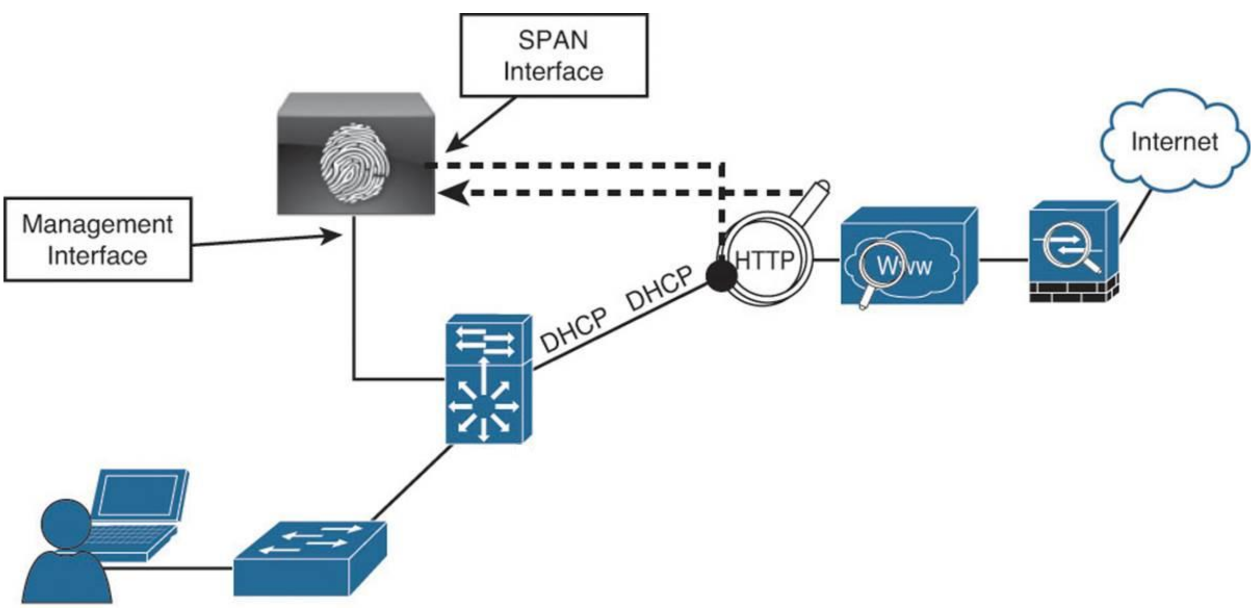




HTTP Probe:

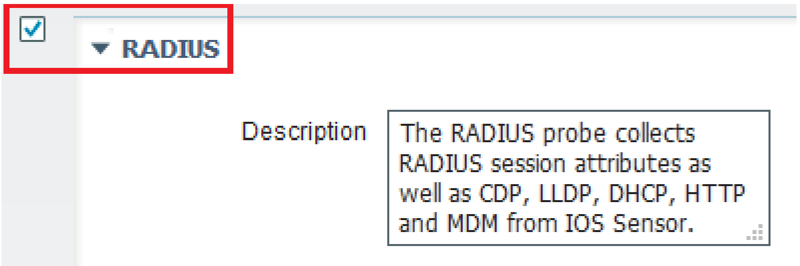
- Information is transmitted in HTTP request-header field called User-Agent field.
- The Cisco Identity Services Engine uses the information in HTTP packets.
- User-Agent field, to help match signatures of what profile a device belongs in.
- Use a Switched Port Analyzer (SPAN) session in true promiscuous mode.





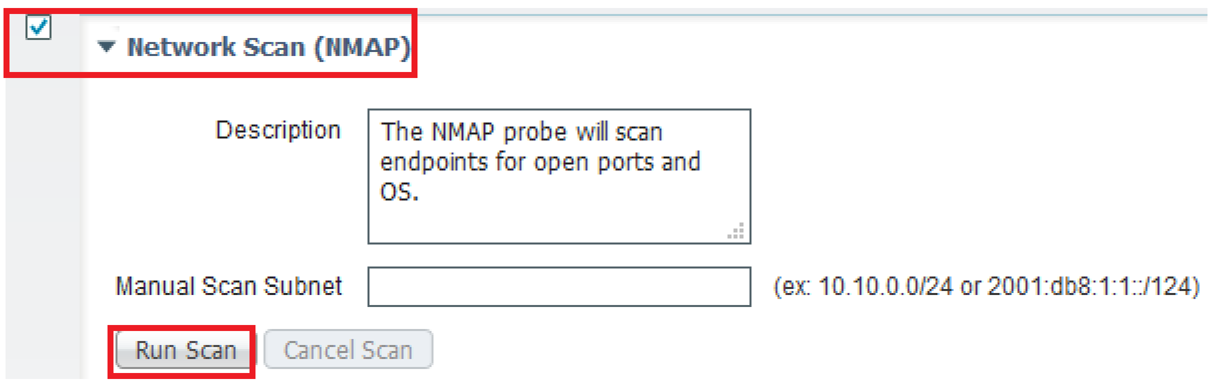
RADIUS Probe:

- Calling-Station-ID field in RADIUS packet provides the endpoint's MAC address.
- Framed-IP-Address field provides its IP address in the RADIUS accounting packet.
- RADIUS probe in Profiling is use to Collect attributes from the RADIUS Attributes.
- RADIUS Probe also collects other information from accounting packet of RADIUS.
- RADIUS Probe also collects other information like CDP, LLDP and DHCP attributes.



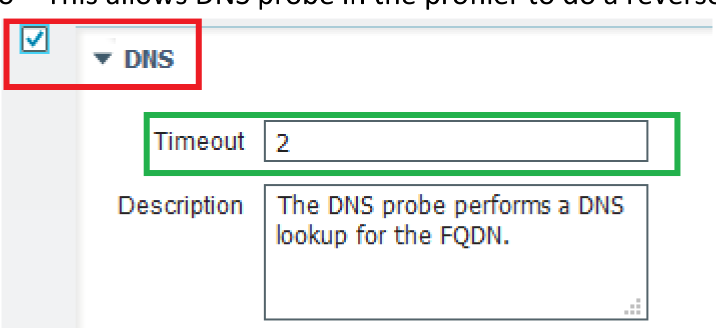
Network Scan (NMAP) Probe:

- Endpoint Scanning (NMAP) probe is executed against an IP Address or subnet.
- NMAP is a tool uses port scans, to identity a device's OS or other attributes of device.
- Also, can run manual scan against a single node, or an entire network.
- NMAP Probe scan endpoints for open ports and Operating System to get information.



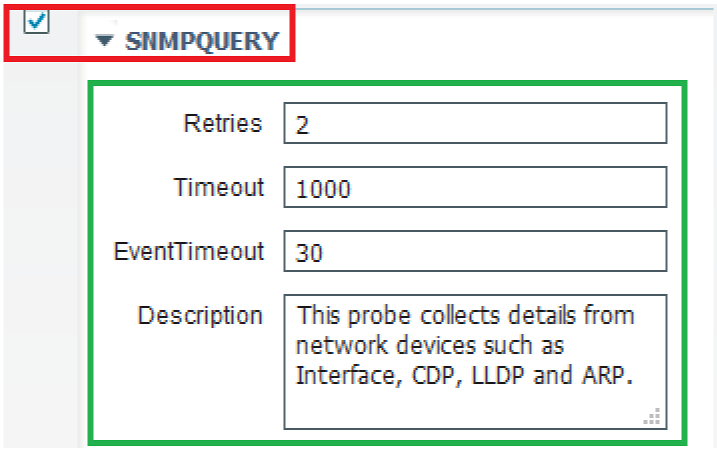
DNS Probe:

- DNS probe is used to collect the fully qualified domain name (FQDN) of endpoint.
- Useful looking for a specific DNS name format of assets Active Directory members.
- DNS probe require anyone from these probe DHCP, DHCP SPAN, HTTP, RADIUS, or SNMP.
- This allows DNS probe in the profiler to do a reverse DNS lookup against specified name.



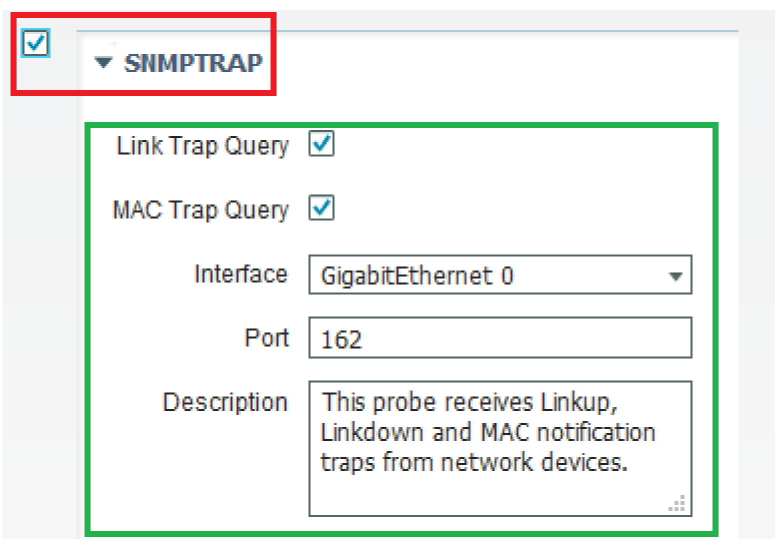
SNMPQUERY Probe:

- SNMP Query Probe send query packets known as SNMP GET Requests.
- System Queries are periodic depending upon polling interval.
- System Queries collect Bridge, IP CDP Cache Entry LLDP Local System Data.
- Interface Queries are generated when RADIUS Accounting Starts.
- Interface Queries are generated when SNMP detect linkup.



SNMP Trap Probe:

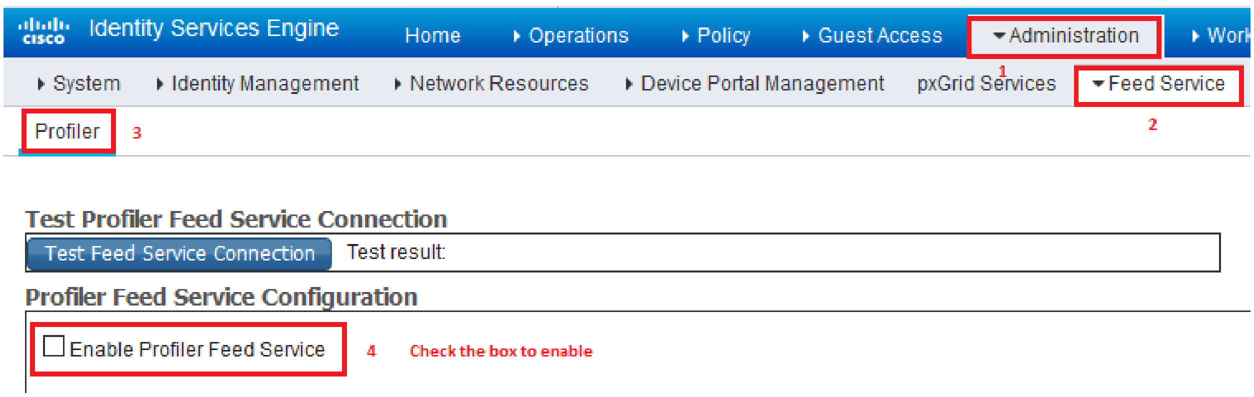
- SNMP Trap receives information from configured NAD support MAC notification.
- SNMP Trap receives info form configured NAD support linkup, linkdown & informs.
- For SNMPTRAP to be functional, you must also enable the SNMPQUERY probe.
- To make this feature functional, configure the NAD to send SNMP traps or informs.
- SNMP Trap probe receives information from the specific network access devices.
- When ports up or go down & endpoints disconnect from or connect to network.



Profiler Feed Service:

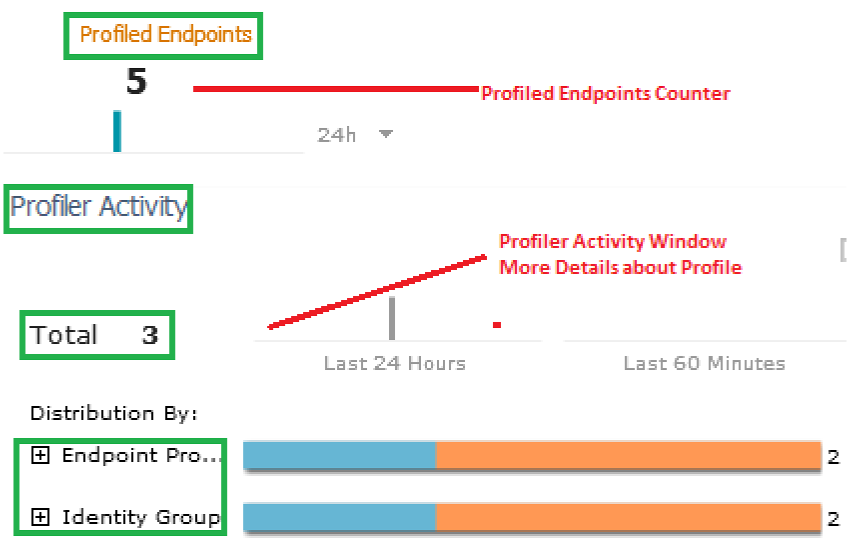
- ISE receive updates for new, updated endpoint profiling policies and MAC OUI.
- Cisco ISE use a feed from a designated Cisco feed server to pull profiling updates.
- Enabling the profiler feed service allows Cisco ISE to check for new & updated profiles.
- The profiler feed service, that you can enable will download new profiles from Cisco.
- By default, profiler feed service is disabled, it requires Advance license to enable it.
- In order to get updated profiles, enable the Profiler Feed Service so get new profiles.

To enable this service, navigate to **Administration>Feed Service>Profiler** and check the box next to **Enable Profiler Feed Service**. Click Save when done:



Verify Profiling:

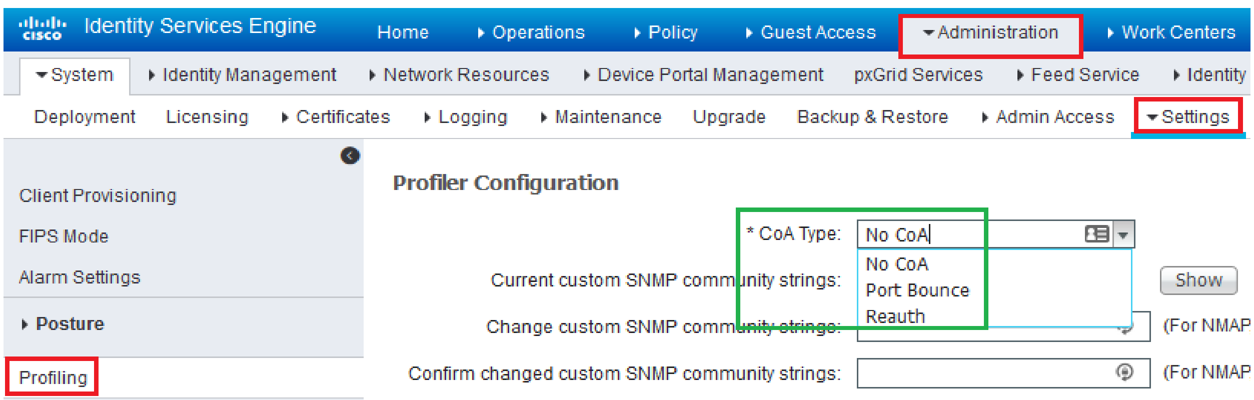
- There are many places to verify profiling operation one of them is dashboard.
- Dashboard are numerous places to look to see whether profiling is working.
- Counter shows number of endpoints that have been profiled in the last 24 hours.
- Profiler Activity Window shows more detail than the Profiled Endpoints Counter.
- Profiler Activity Window shows two different counters last 60 minutes & 24 hours.



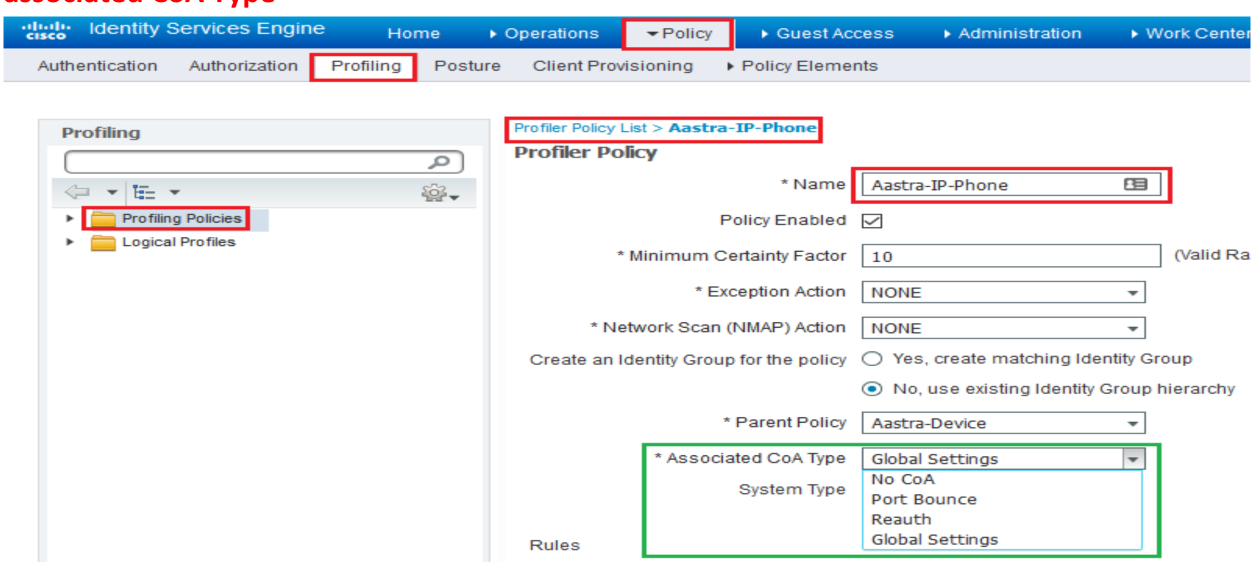
Implement CoA:

- o Profiling requires CoA to change the authorization state of an endpoint.
- o Cisco ISE allows a global configuration to issue a Change of Authorization (CoA).
- o Enables profiling service with more control over endpoints already authenticated.
- o After profiler CoA is enabled globally, CoA is automatically sent for any endpoint.
- o CoA sent for any endpoint that transitions from unknown to any known profile.
- o No CoA (default)-Use this option to disable the global configuration of CoA.
- o Port Bounce-Use this option, if the switch port exists with only one session.
- o Port Bounce CoA shuts down switch port & performs no shutdown to reenable it.
- o Reauth-Use this option to enforce reauthentication of already authenticated endpoint.
- o If the port exists with multiple sessions, then use the Reauth CoA option.
- o Change of Authorization (CoA) can be enable Per-profile as well.

To modify or change this service, navigate to **Administration>Setting>Profiling**



To modify per-profile, navigate to **Policy>Profiling>Profiling Policies** Click on any profile there is **associated CoA Type**

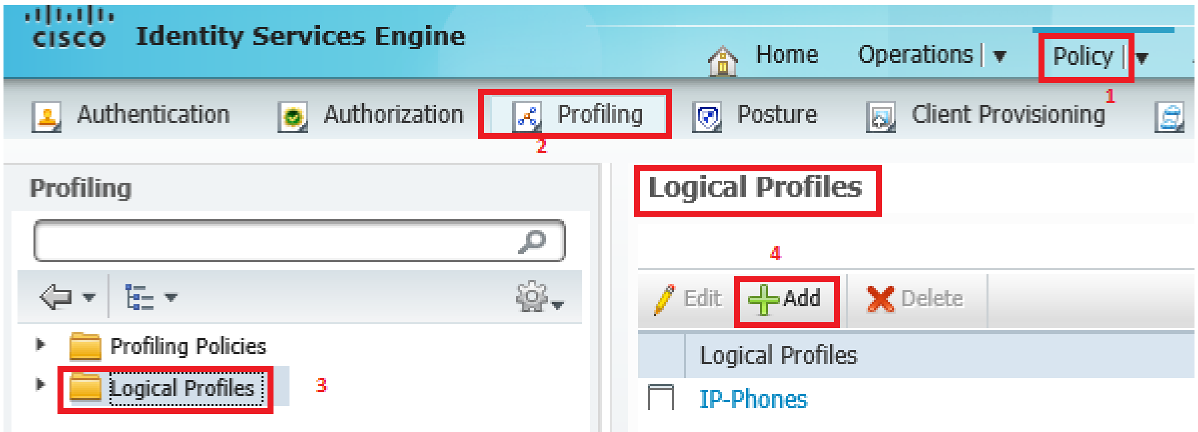


 Prepared By Ahmad Ali, Email: ahmadalimsc@gmail.com , Mobile# 0564303717

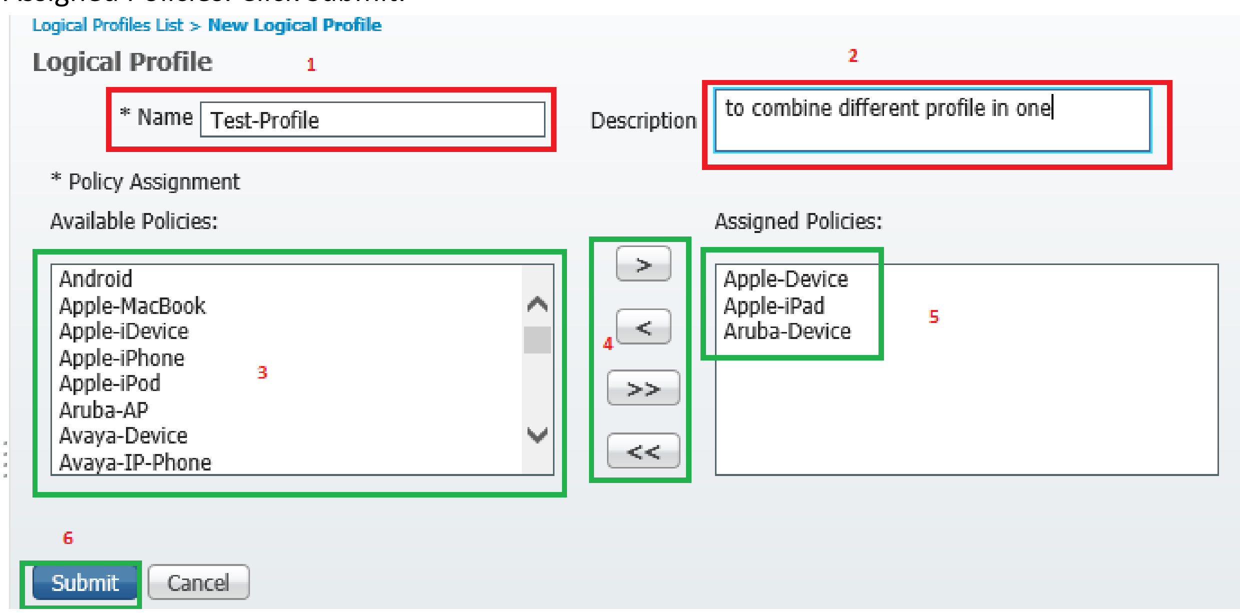
Logical Profile:

- Logical profile is a container for a category of profiles or associated profiles.
- Logical profile administrator-created endpoint profiling groups and policies.
- An endpoint profiling policy can be associated to multiple logical profiles.
- Logical profile can be used in an authorization policy condition as well.
- Logical profiles allow to group different type of dynamically profiled devices.

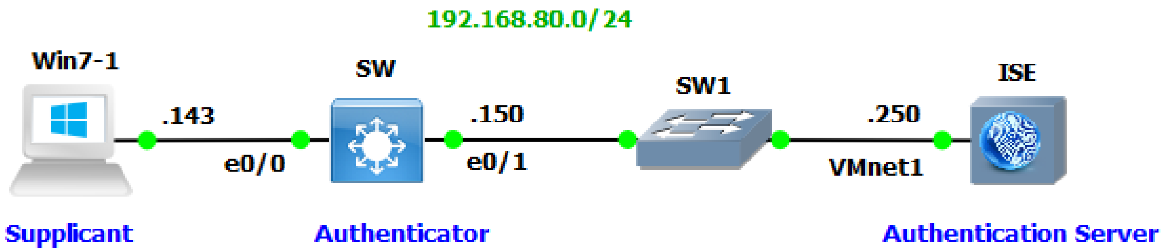
Choose Policy > Profiling > Logical Profiles.



Click Add. Enter a name and description for the new logical profile in the text boxes for Name and Description. Choose endpoint-profiling policies from the Available Policies to assign them in a logical profile. Click the right arrow to move the selected endpoint profiling policies to the Assigned Policies. Click Submit.



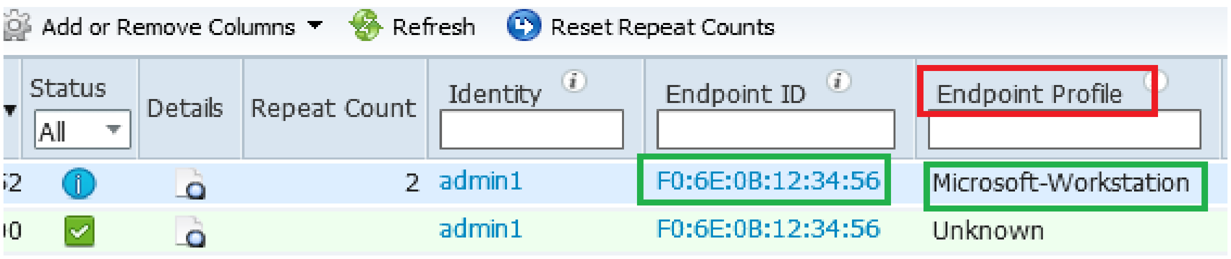
Lab Topology:

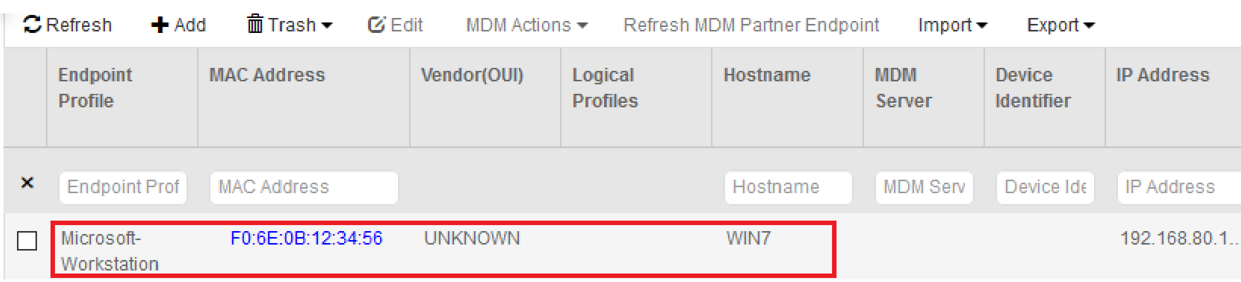


Dot1x Configuration
SW(config)#interface vlan 1 SW(config-if)#ip add 192.168.80.150 255.255.255.0 SW(config-if)#no shutdown SW(config-if)#exit
SW(config)# aaa new-model
SW(config)# username admin password 123
SW(config)# radius server myserver SW(config-radius-server)#address ipv4 192.168.80.250 SW(config-radius-server)#key test123
SW(config)# aaa authentication dot1x default group radius
SW(config)#aaa authorization network default group radius
SW(config)#aaa accounting dot1x default start-stop group radius
SW(config)#radius-server vsa send
SW(config)#radius-server vsa send accounting
SW(config)#radius-server vsa send authentication
SW(config)#radius-server attribute 6 on-for-login-auth SW(config)#radius-server attribute 8 include-in-access-req SW(config)#radius-server attribute 25 access-request include
SW(config)#ip device tracking
SW(config)# dot1x system-auth-control
SW(config)# interface e0/0 SW(config-if)# switchport host SW(config-if)# switchport access vlan 1 SW(config-if)# authentication port-control auto SW(config-if)# dot1x pae authenticator SW(config-if)# authentication host-mode multi-host SW(config-if)#authentication timer reauthenticate server SW(config-if)#authentication event fail action next-method SW(config-if)#authentication periodic SW(config-if)#authentication violation shutdown
SW# show dot1x interface e0/0
SW# show dot1x all

RADIUS Probe:

- RADIUS Probe is one of the most common probes using by ISE.
- ISE profile based on RADIUS attributes collected from RADIUS messages.
- RADIUS probe also listens to CDP & DHCP attributes send in RADIUS accounting packets.
- Such as User-Name, Calling-Station-Id, NAS-IP-Address, NAS-Port & Framed-IP-Address.
- There is a lot of more information that can be pulled from the RADIUS attributes.





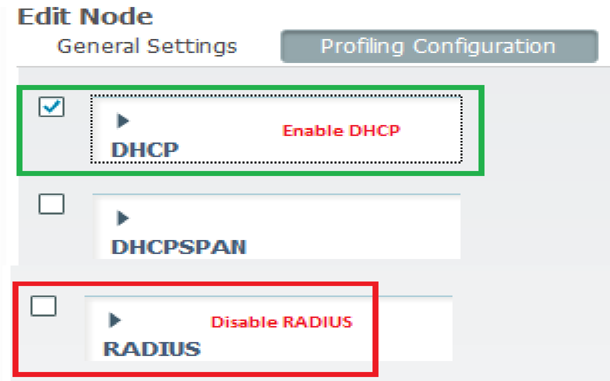
Looking at one of profiled hosts **Administration>Identity Management> Identities> Endpoints**, see the attributes that were collected:

Called-Station-ID	AA-BB-CC-00-01-00
Calling-Station-ID	F0-6E-0B-12-34-56
DestinationIPAddress	192.168.80.250
DestinationPort	1645
Device IP Address	192.168.80.150
Device Type	Device Type#All Device Types
DeviceRegistrationStatus	NotRegistered
ElapsedDays	0
EndPointMACAddress	F0-6E-0B-12-34-56
EndPointPolicy	Microsoft-Workstation
EndPointProfilerServer	ISE1.google.com
EndPointSource	RADIUS Probe
Framed-IP-Address	192.168.80.148
IdentityGroup	Workstation

DHCP Probe:

- DHCP Probe collect DHCP request attribute from user, proxy & Helper.
- Traffic forwarded to Cisco ISE with helper address or SPAN.
- DHCP Probe using ip helper-address interface configuration command.
- Cisco ISE will only use the incoming DHCP data to profile endpoints.

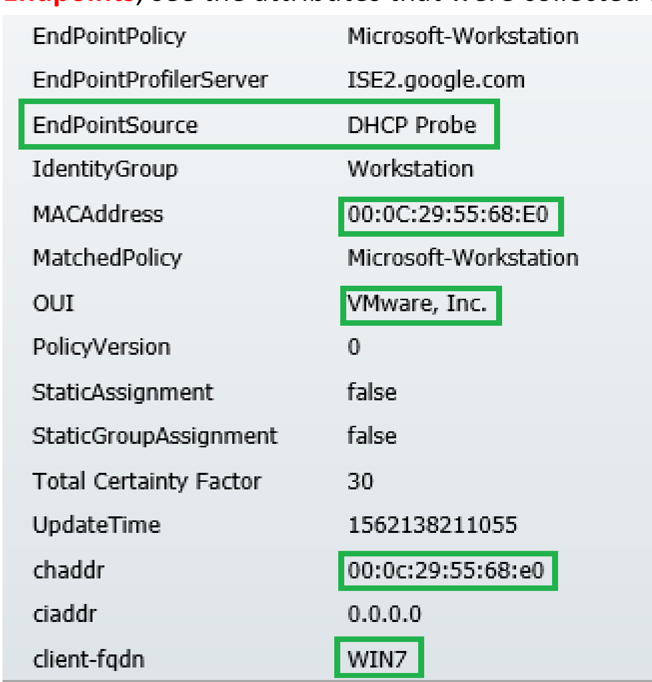
Navigate to **Administration >System >Deployment > Profiling Configuration** this time disable RADIUS Probe and only enable DHCP Probe.



Configure the switch to forward DHCP request to Cisco ISE.

```
SW(config)#interface vlan 1
SW(config-if)#ip helper-address 192.168.80.1
SW(config-if)#ip helper-address 192.168.80.250
```

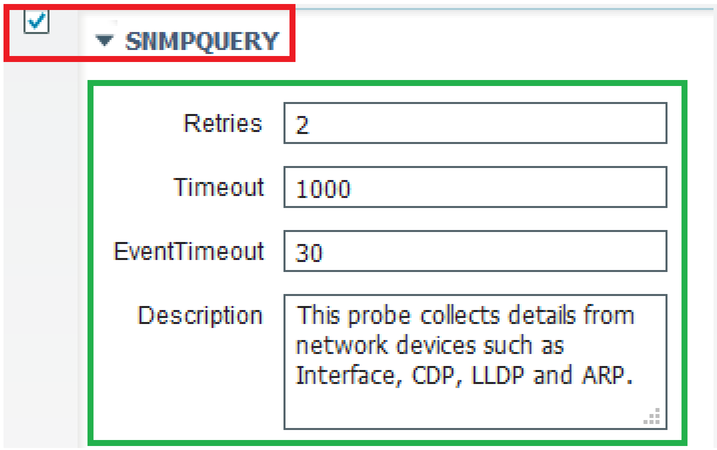
Let us look again the profiled hosts **Administration>Identity Management> Identities> Endpoints**, see the attributes that were collected this time by DHCP.



EndPointPolicy	Microsoft-Workstation
EndPointProfilerServer	ISE2.google.com
EndPointSource	DHCP Probe
IdentityGroup	Workstation
MACAddress	00:0C:29:55:68:E0
MatchedPolicy	Microsoft-Workstation
OUI	VMware, Inc.
PolicyVersion	0
StaticAssignment	false
StaticGroupAssignment	false
Total Certainty Factor	30
UpdateTime	1562138211055
chaddr	00:0c:29:55:68:e0
ciaddr	0.0.0.0
client-fqdn	WIN7

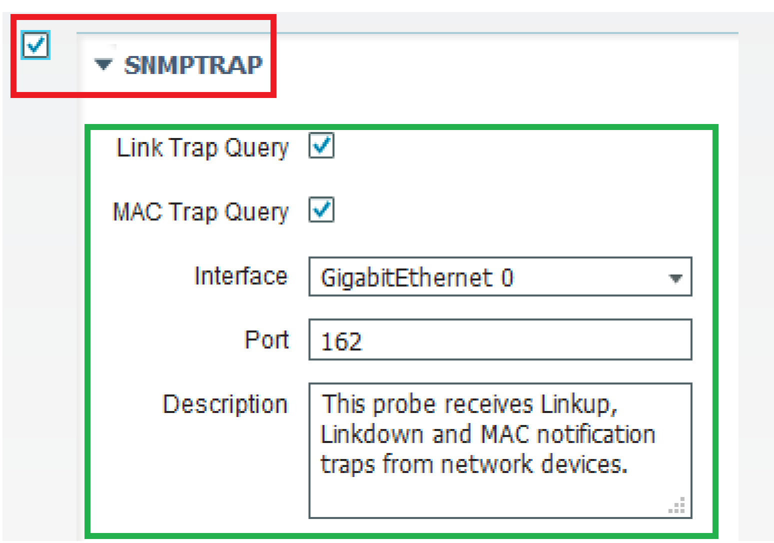
SNMPQUERY Probe:

- SNMP Query Probe send query packets known as SNMP GET Requests.
- System Queries are periodic depending upon polling interval.
- System Queries collect Bridge, IP CDP Cache Entry LLDP Local System Data.
- Interface Queries are generated when RADIUS Accounting Starts.
- Interface Queries are generated when SNMP detect linkup.

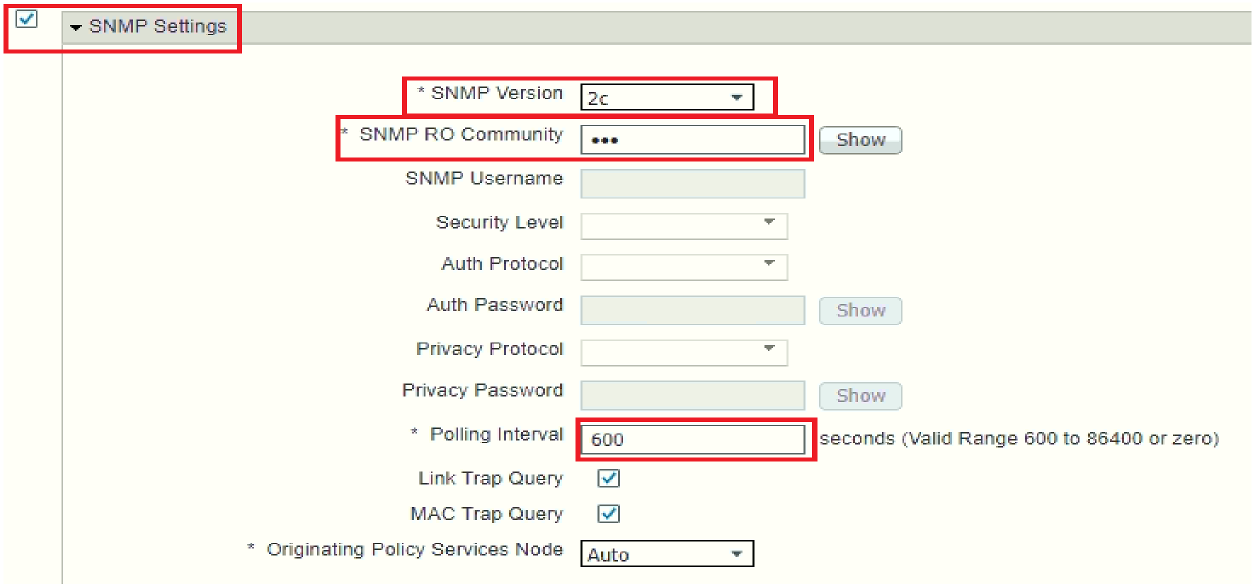


SNMP Trap Probe:

- SNMP Trap receives information from configured NAD support MAC notification.
- SNMP Trap receives info form configured NAD support linkup, linkdown & informs.
- For SNMPTRAP to be functional, you must also enable the SNMPQUERY probe.
- To make this feature functional, configure the NAD to send SNMP traps or informs.
- SNMP Trap probe receives information from the specific network access devices.
- When ports up or go down & endpoints disconnect from or connect to network.



Go to **Administration > Network Resources > Network devices** click on network device in my case switch go down to **SNMP Settings** enable **SNMP Settings**, Choose version **2c** and type SNMP Community **123**. Click **Save** button to save the settings.



Configuration of SNMP on Switch
S1(configure)# snmp-server community **123** rw
S1(configure)# snmp-server enable traps
S1(configure)# snmp-server host 192.168.8.250 **123**

Let us look again the profiled hosts **Administration>Identity Management> Identities> Endpoints**, see the attributes that were collected this time by **SNMPQuary**.